

Software Performance Engineering

Recitation 2.1

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Some due dates

- Team contract of project 2 is due next Tuesday, Oct 28.
- One group only need to submit 1 copy, including all team members' names.
- Submit on GradeScope and include it in your GitHub repo.
- It's ok to update your team contract later, as long as all members agree on it.
-
- Homework 5 will release next Thursday, Oct 30.

PROJECT 2: RENDERING AND SIMULATING 3D SCENES



Check Off

Clone your project repo and run:

```
1  make clean; make
2  ./bin/ref-test tiers/1/s tiers/1/r
3  ./bin/find-tier
```

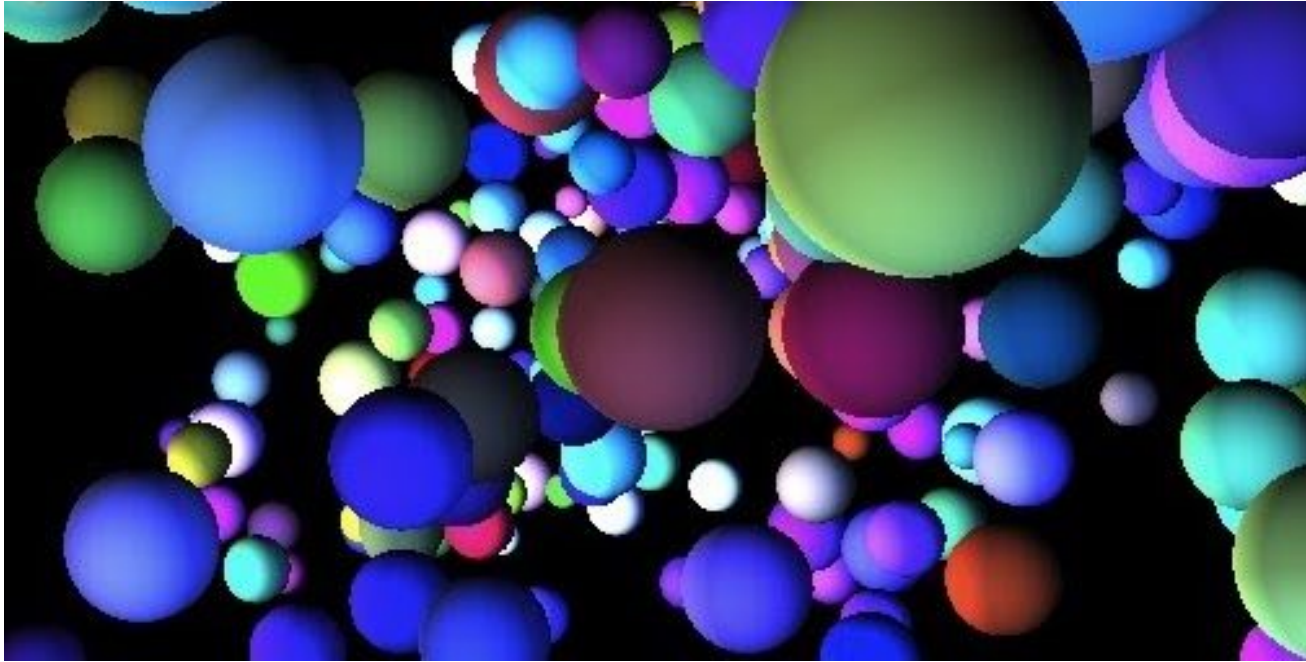
The first is the correctness test, and the second is the performance test.

Example Output

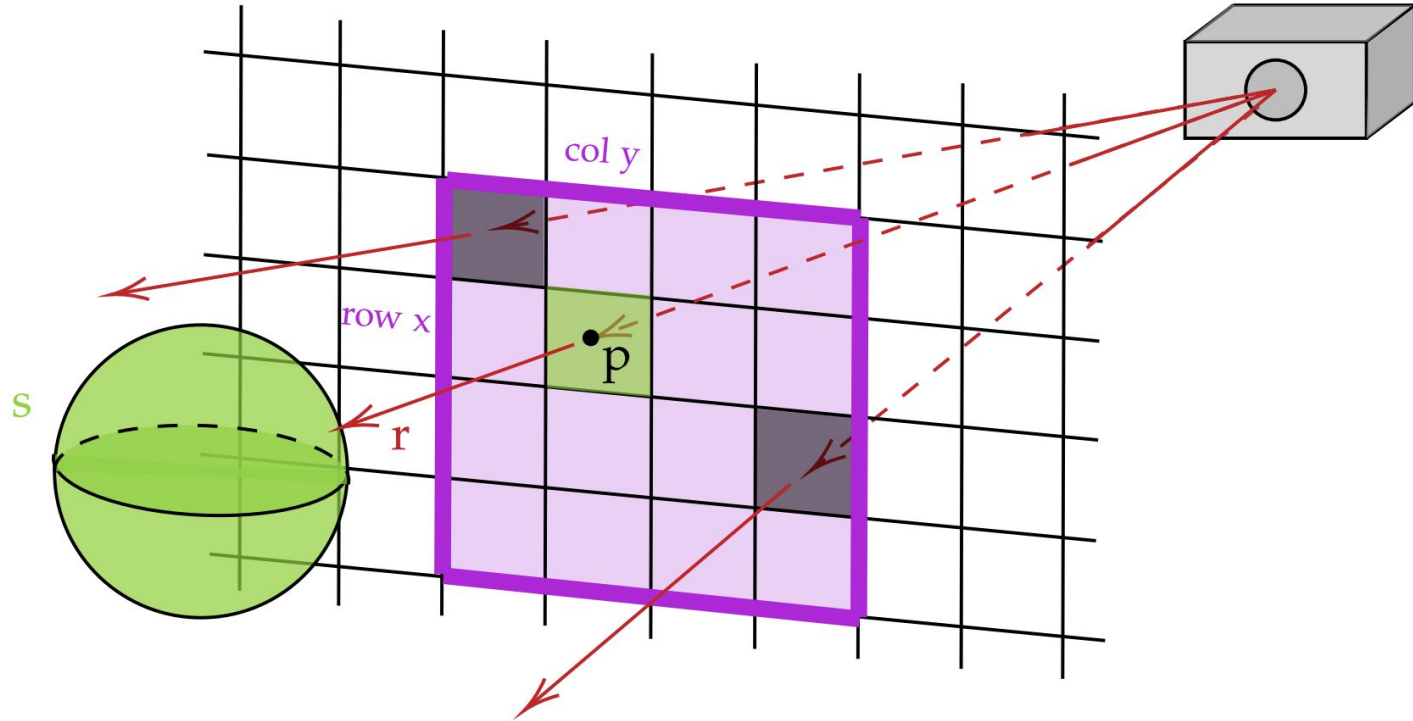
```
ref-test options:  
  s_spec = tiers/1/s  
  r_spec = tiers/1/r  
  renderer = student  
  simulator = student  
  n_frames = 12  
Average difference: 0.0000000000000000  
Standard deviation: 0.0000000000000000  
Minimum difference: 0.0000000000000000  
Maximum difference: 0.0000000000000000  
Test result: PASS
```

```
PASS (boyah!): Tier 39 : Ran 1728x1728 image with 3093 bodies in 321ms (s: 101, r: 100, is: 0, ir: 0)  
PASS (woot!): Tier 40 : Ran 1536x1536 image with 3840 bodies in 388ms (s: 176, r: 211, is: 0, ir: 0)  
PASS (ayy!): Tier 41 : Ran 1584x1584 image with 3994 bodies in 428ms (s: 190, r: 237, is: 0, ir: 0)  
PASS (yeee!): Tier 42 : Ran 1648x1648 image with 4154 bodies in 431ms (s: 203, r: 225, is: 0, ir: 0)  
PASS (yay!): Tier 43 : Ran 1728x1728 image with 4320 bodies in 497ms (s: 224, r: 269, is: 0, ir: 0)  
PASS (eoo!): Tier 44 : Ran 1792x1792 image with 4493 bodies in 540ms (s: 240, r: 298, is: 0, ir: 0)  
PASS (skrrt!): Tier 45 : Ran 1856x1856 image with 4672 bodies in 601ms (s: 261, r: 337, is: 0, ir: 0)  
PASS (eoo!): Tier 46 : Ran 1936x1936 image with 4859 bodies in 598ms (s: 280, r: 315, is: 0, ir: 0)  
PASS (boyah!): Tier 47 : Ran 2016x2016 image with 5054 bodies in 678ms (s: 304, r: 371, is: 0, ir: 0)  
PASS (boyah!): Tier 48 : Ran 2096x2096 image with 5256 bodies in 722ms (s: 333, r: 386, is: 0, ir: 0)  
PASS (eoo!): Tier 49 : Ran 2176x2176 image with 5466 bodies in 800ms (s: 358, r: 524, is: 0, ir: 0)
```

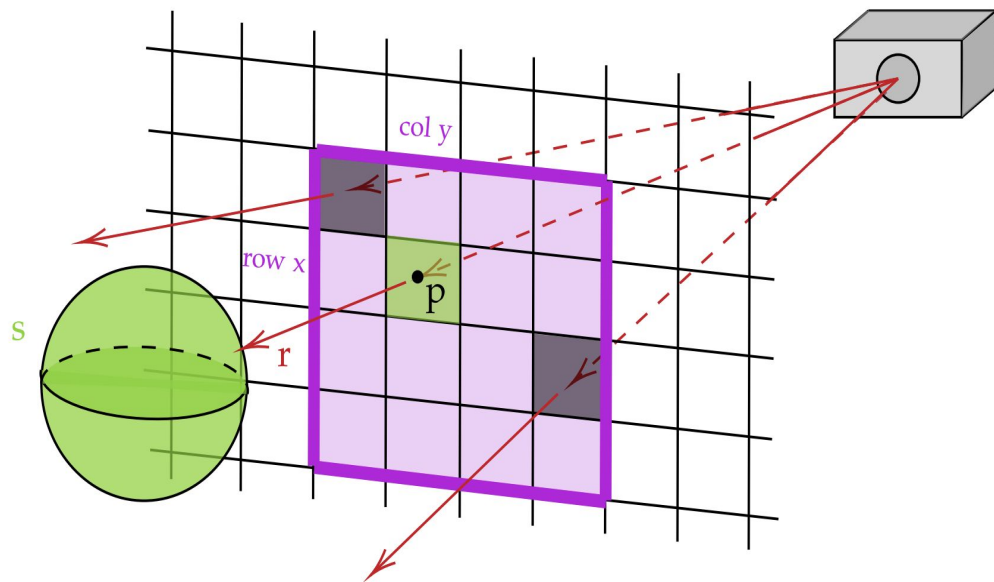
Project 2: N body simulation



Ray Tracing

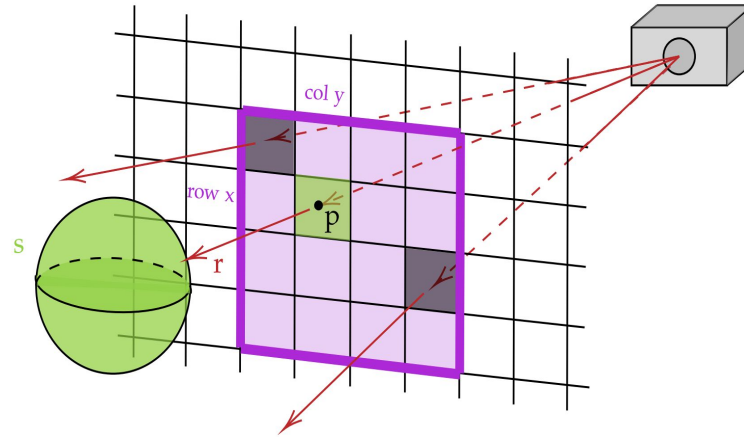


Elements of Ray Tracing



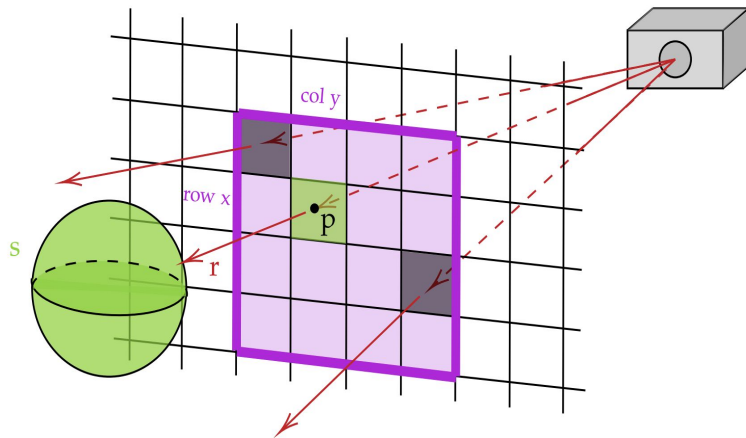
1. Origin (O)
2. Eye location (e)
3. Image Plane (can be inclined to the eye)
4. Plane orthonormal vectors (u and v)
5. Sphere : radius and center
6. Output Image is a resolution x resolution (square) image of pixels
7. Pixels have a fixed width, given by viewport/resolution

3 coordinates (2 spaces)



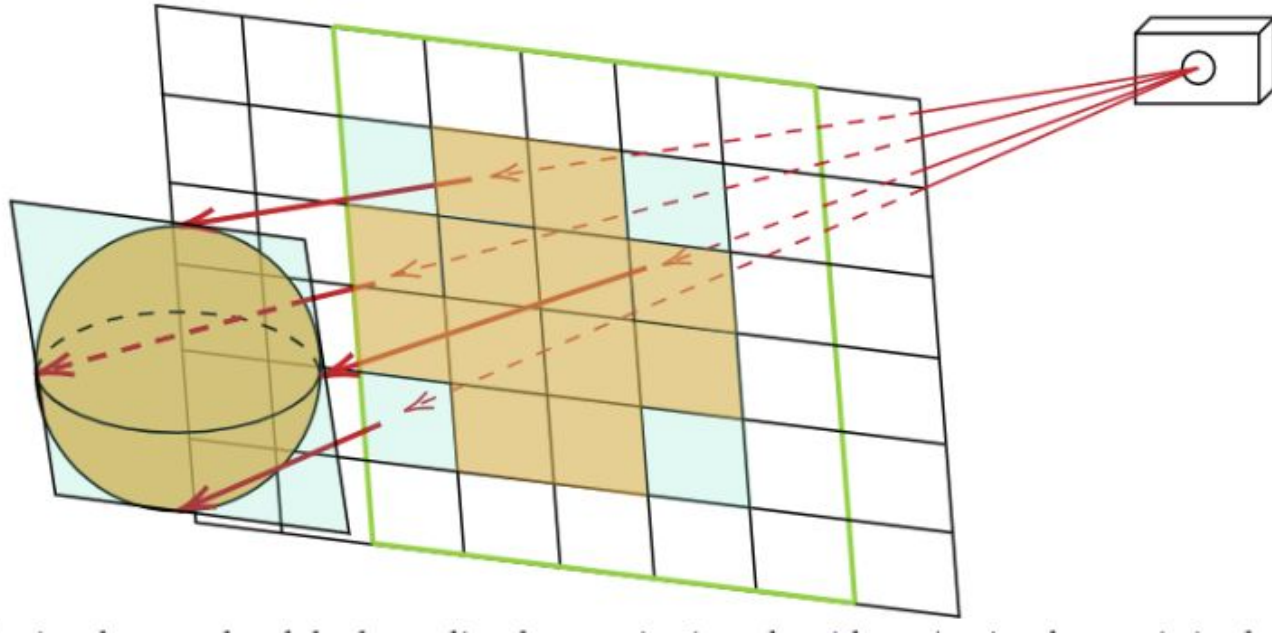
- X, Y, Z of the spheres
- X, Y, Z of the coordinates occurring on the plane (same units as spheres)
- I, J of the image (labeled x, y in the code)

How do we get the pixel value at I and J?

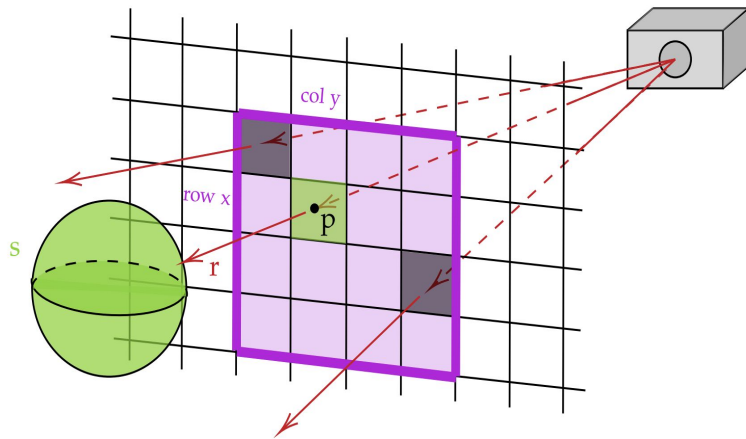


- Given an image pixel location (I, J)
- Find the point in the 3D plane that corresponds to this pixel location (calculate ray from the eye)
- Determine if the ray hits a sphere
 - Complicated math

Bounding Boxes



A different way?

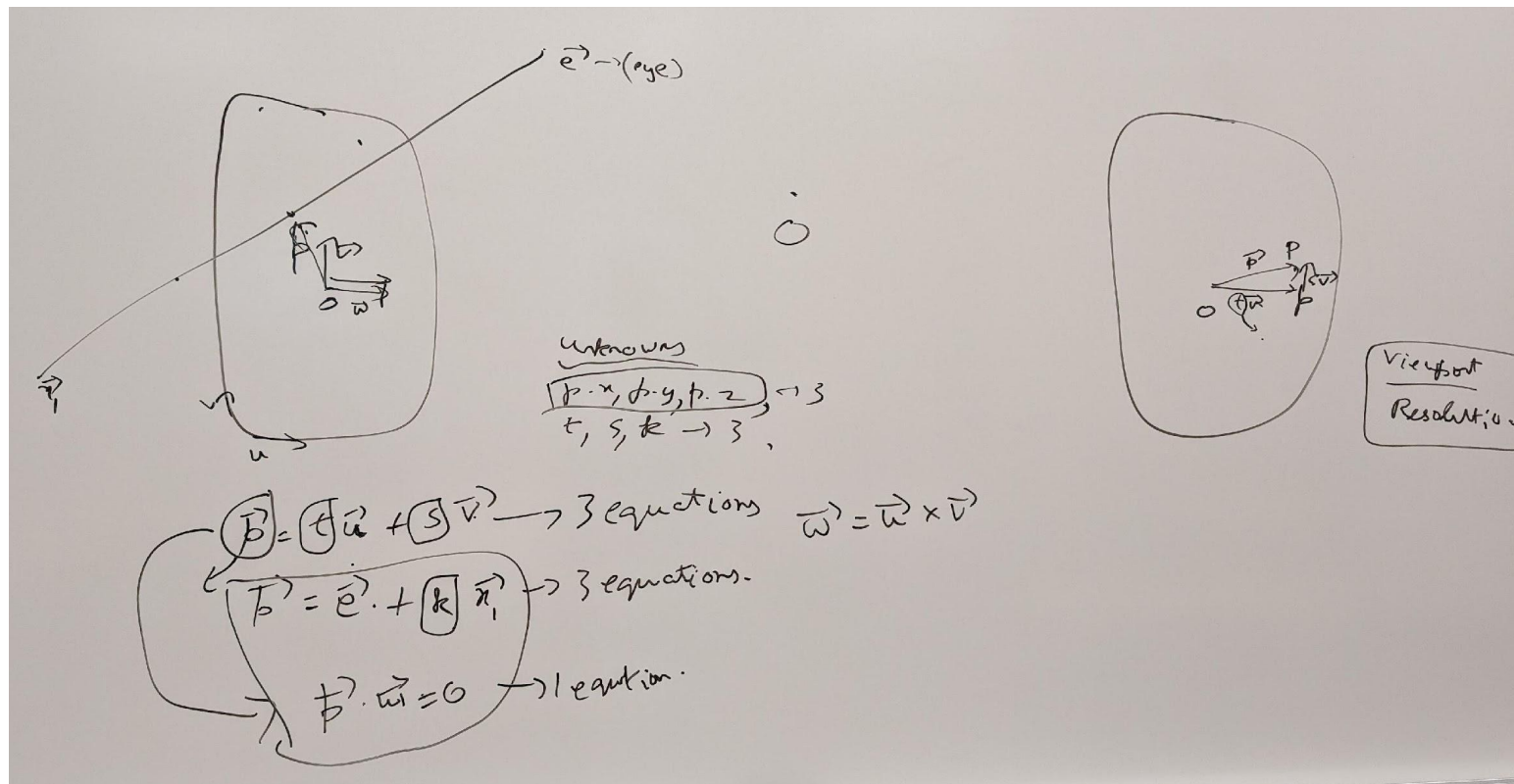


- Given a sphere location, get 8 points making a bounding cube
 - Ex: Given r and (x, y, z) some corners are $(x+r, y+r, z+r)$, $(x-r, y-r, z-r)$, $(x+r, y-r, z-r)$
- Project those points onto the plane
- For the bounding box of the cube in the image, check if the points intersect the sphere
 - If so, color the pixel

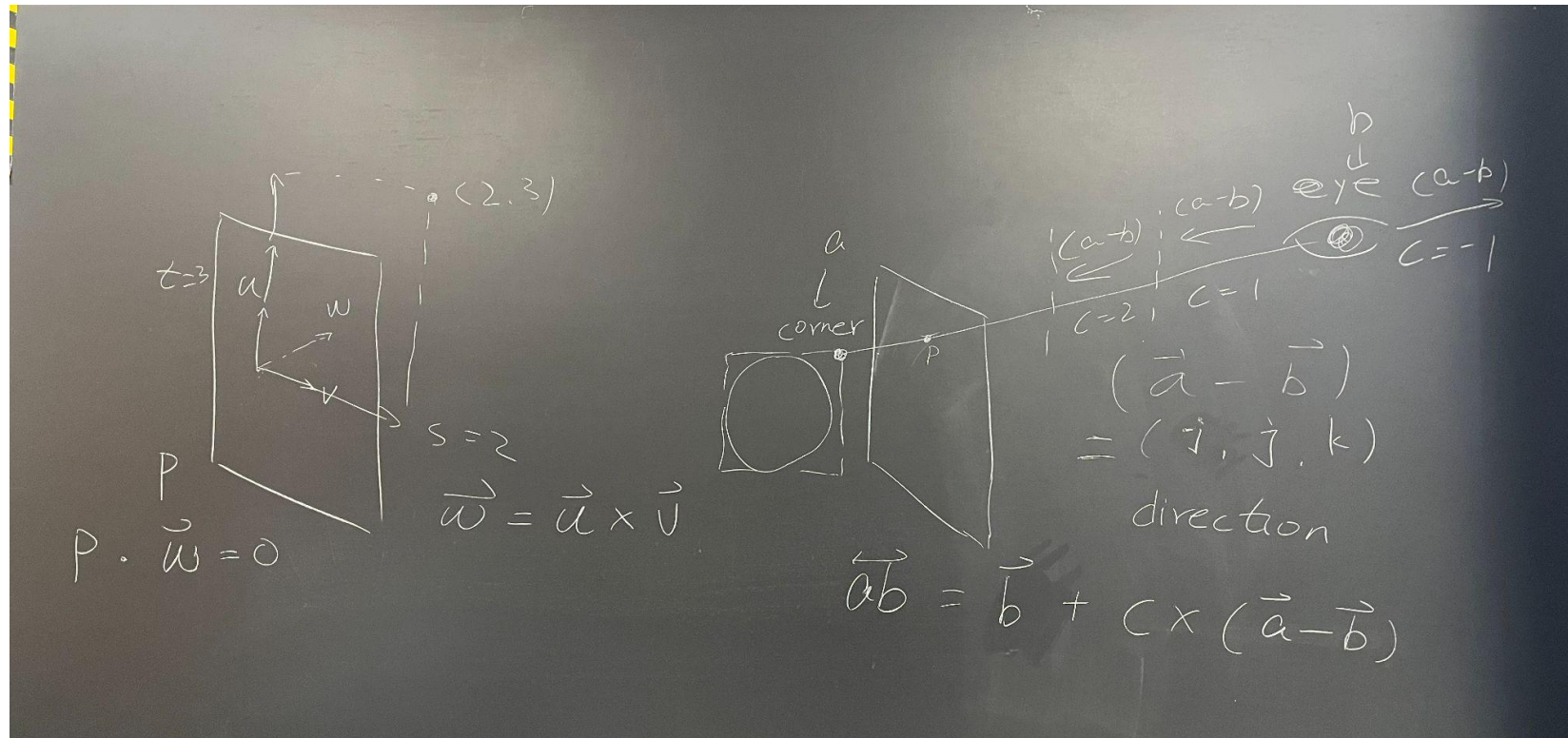
Project the cube onto the plane

- Given points a, b find where they intersect a plane through the origin with orthonormal vectors u, v
- $\overrightarrow{ab} = \vec{b} + c * (\vec{a} - \vec{b})$ for any number c
- Plane = $t\vec{u} + s\vec{v}$ for any numbers t, s
 - Also given by $p \cdot \vec{w} = 0$ where $\vec{w} = \vec{u} \times \vec{v}$
- 3 unknowns, 3 equations (one each for x, y, z components)
- Solve to get the 8 points and get a bounding box of those 8 points
- You can then have a bigger bounding box with just 4 points

Vector Algebra for Rendering



Vector Algebra for Rendering



CODING TIME!

